

10th grade

Learning evidence 2.1: Analysis of Decisions (Solutions)

Problems

Problem 1: Problem description

A school cafeteria must choose one of three menu plans for a two-week festival, Plan A (local produce focus), Plan B (mixed menu), or Plan C (bulk staples). Demand for meals can be very high, high, medium, or low. Based on past festivals, the probability of very high demand is 0,15, high demand is 0,30, medium demand is 0,35, and low demand is 0,20. Profits are measured in hundreds of dollars. If demand is very high, Plan A yields 105, Plan B yields 85, and Plan C yields 70. If demand is high, Plan A yields 92, Plan B yields 70, and Plan C yields 55. If demand is medium, Plan A yields 50, Plan B yields 62, and Plan C yields 48. If demand is low, Plan A yields –8, Plan B yields 30, and Plan C yields 38. The stated probabilities must be used for an expected value comparison as part of the decision analysis.

C2

Decision Alternatives	Plan A (local produce), Plan B (mixed menu), Plan C (bulk staples)
States of Nature	Demand very high, high, medium, low with probabilities 0,15, 0,30, 0,35, 0,20
Events	Realized demand level in the serving period
Consequences	Profit in hundreds of dollars after all costs

C3

Payoff = revenue – cost. In this problem, the profits for each plan are already provided and directly represent the corresponding payoffs.

<i>Payoff table</i>				
State of nature	Probability	Plan A	Plan B	Plan C
Very high demand	0.15	105	85	70
High demand	0.30	92	70	55
Medium demand	0.35	50	62	48
Low demand	0.20	-8	30	38

C4

$$\text{máx}(\text{Plan A}) = \text{máx}\{105, 92, 50, -8\} = 105,$$

$$\text{máx}(\text{Plan B}) = \text{máx}\{85, 70, 62, 30\} = 85,$$

$$\text{máx}(\text{Plan C}) = \text{máx}\{70, 55, 48, 38\} = 70.$$

$$\text{máx}\{\text{máx}(\text{Plan A}), \text{máx}(\text{Plan B}), \text{máx}(\text{Plan C})\} = \text{máx}\{105, 85, 70\} = 105.$$

The Maximax choice is Plan A. It is *risky* because it only considers the highest possible profit.

C5

$$\min(\text{Plan A}) = \min\{105, 92, 50, -8\} = -8,$$

$$\min(\text{Plan B}) = \min\{85, 70, 62, 30\} = 30,$$

$$\min(\text{Plan C}) = \min\{70, 55, 48, 38\} = 38.$$

$$\max\{\min(\text{Plan A}), \min(\text{Plan B}), \min(\text{Plan C})\} = \max\{-8, 30, 38\} = 38.$$

The Maximin choice is Plan C with 38 because it gives the best worst-case outcome. It is *safe* because it prioritizes the least harmful result.

C1

$$EV_A = 0,15(105) + 0,30(92) + 0,35(50) + 0,20(-8) = 15,75 + 27,6 + 17,5 - 1,6 = 59,25,$$

$$EV_B = 0,15(85) + 0,30(70) + 0,35(62) + 0,20(30) = 12,75 + 21 + 21,7 + 6 = 61,45,$$

$$EV_C = 0,15(70) + 0,30(55) + 0,35(48) + 0,20(38) = 10,5 + 16,5 + 16,8 + 7,6 = 51,4.$$

The expected value criterion chooses Plan B because 61,45 is the largest expected profit. This differs from the Maximax choice Plan A and the Maximin choice Plan C, so the expected value gives a balanced decision using the probabilities.

Problem 2: Problem description

A delivery cooperative is choosing a fleet strategy for the next season, Strategy A (electric vans) or Strategy B (hybrid vans). Demand for deliveries can be strong or weak, and fuel costs can be low or high. The cooperative estimates the demand probabilities as 0,60 (strong) and 0,40 (weak). Fuel costs are independent of demand, with probabilities 0,55 (low) and 0,45 (high). Revenue, in thousands of dollars, depends on the demand level and strategy. Under strong demand, revenue is 420 for Strategy A and 460 for Strategy B. Under weak demand, revenue is 210 for Strategy A and 230 for Strategy B. Operating costs, in thousands of dollars, depend on the fuel cost level and strategy. If fuel costs are low, operating costs are 160 for Strategy A and 190 for Strategy B. If fuel costs are high, operating costs are 220 for Strategy A and 250 for Strategy B. The cooperative should treat each combined demand and fuel-cost outcome as a state of nature and use the stated probabilities to compute expected profits.

C2

Decision Alternatives	Strategy A (electric), Strategy B (hybrid)
States of Nature	Demand strong, weak with fuel costs low or high
Events	Realized demand level paired with fuel cost conditions in the season
Consequences	Profit in thousands of dollars from revenue minus operating costs

C3

Payoff = revenue – cost, where revenue is determined by demand and cost is determined by fuel prices.

Revenue parameters by strategy

Strategy	Strong	Weak
Probabilities	0.60	0.40
Strategy A	420	210
Strategy B	460	230

Cost parameters by strategy

Strategy	Low cost	High cost
Probabilities	0.55	0.45
Strategy A	160	220
Strategy B	190	250

Profit is computed as Revenue minus Cost for each alternative and state.

Profit parameters by strategy

State of nature	Probability	Strategy A	Strategy B
Strong demand, low cost	$0,60 \cdot 0,55$ $= 0,33$	$420 - 160$ $= 260$	$460 - 190$ $= 270$
Strong demand, high cost	$0,60 \cdot 0,45$ $= 0,27$	$420 - 220$ $= 200$	$460 - 250$ $= 210$
Weak demand, low cost	$0,40 \cdot 0,55$ $= 0,22$	$210 - 160$ $= 50$	$230 - 190$ $= 40$
Weak demand, high cost	$0,40 \cdot 0,45$ $= 0,18$	$210 - 220$ $= -10$	$230 - 250$ $= -20$

Payoff table

State of nature	Probability	Strategy A	Strategy B
Strong demand, low cost	0.33	260	270
Strong demand, high cost	0.27	200	210
Weak demand, low cost	0.22	50	40
Weak demand, high cost	0.18	-10	-20

C4

$$\text{máx}(\text{Strategy A}) = \text{máx}\{260, 200, 50, -10\} = 260,$$

$$\text{máx}(\text{Strategy B}) = \text{máx}\{270, 210, 40, -20\} = 270.$$

$$\text{máx}\{\text{máx}(\text{Strategy A}), \text{máx}(\text{Strategy B})\} = \text{máx}\{260, 270\} = 270.$$

The Maximax choice is Strategy B. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\text{mín}(\text{Strategy A}) = \text{mín}\{260, 200, 50, -10\} = -10,$$

$$\text{mín}(\text{Strategy B}) = \text{mín}\{270, 210, 40, -20\} = -20.$$

$$\text{máx}\{\text{mín}(\text{Strategy A}), \text{mín}(\text{Strategy B})\} = \text{máx}\{-10, -20\} = -10.$$

The Maximin choice is Strategy A with -10 because it has the least severe worst-case loss. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined-state probabilities.

$$\begin{aligned}EV_A &= 0,33(260) + 0,27(200) + 0,22(50) + 0,18(-10) \\&= 85,8 + 54 + 11 - 1,8 = 149,0, \\EV_B &= 0,33(270) + 0,27(210) + 0,22(40) + 0,18(-20) \\&= 89,1 + 56,7 + 8,8 - 3,6 = 151,0.\end{aligned}$$

The expected value criterion chooses Strategy B because 151,0 is the largest expected profit. This matches the Maximax choice and differs from the Maximin choice, so the expected value emphasizes the weighted outcomes.

Problem 3: Problem description

A community sports center must choose a membership model for the next year, Model A (standard monthly pass) or Model B (flex pass). Demand for memberships can be high or low, and staffing costs can be favorable or unfavorable. Demand probabilities are 0,50 (high) and 0,50 (low). Staffing costs are independent of demand, with probabilities 0,60 (favorable) and 0,40 (unfavorable). Expected membership counts depend on demand and model. Under high demand the center expects 620 members with Model A and 700 with Model B. Under low demand it expects 300 with Model A and 360 with Model B. Membership fees are 35 for Model A and 40 for Model B per member per year. Variable staffing cost per member is 18 when costs are favorable and 26 when costs are unfavorable. All monetary amounts in this point can be treated in dollars. Use the combined demand and staffing outcomes as states of nature and the stated probabilities to compute expected profits.

C2

Decision Alternatives	Model A (standard), Model B (flex)
States of Nature	Demand high, low with staffing costs favorable or unfavorable
Events	Realized demand level paired with staffing cost conditions in the year
Consequences	Annual profit in dollars from membership fees minus staffing costs

C3

Payoff = revenue – cost, where revenue = (members × fee) and cost = (members × variable cost).

Members per model by state of nature

Model	High	Low
Model A	620	300
Model B	700	360

Cost parameters by member

Model	Favorable	Unfavorable
Model A	12	25
Model B	18	26

Revenue parameters by member

Model	High	Low
Model A	36	36
Model B	38	38

<i>Revenue parameters by model</i>		
State of nature	Model A	Model B
High demand	$620 \cdot 36$	$700 \cdot 38$
	$= 22320$	$= 26600$
Low demand	$300 \cdot 36$	$360 \cdot 38$
	$= 10800$	$= 13680$

<i>Costs parameters by model</i>		
State of nature	Model A	Model B
High demand, favorable cost	$620 \cdot 12$	$700 \cdot 18$
	$= 7440$	$= 12600$
High demand, unfavorable cost	$620 \cdot 25$	$700 \cdot 26$
	$= 15500$	$= 18200$
Low demand, favorable cost	$300 \cdot 12$	$360 \cdot 18$
	$= 3600$	$= 6480$
Low demand, unfavorable cost	$300 \cdot 25$	$360 \cdot 26$
	$= 7500$	$= 9360$

Profit is computed as Revenue minus Cost for each alternative and state.

<i>Profit parameters by model</i>			
State of nature	Model A	Model B	Probabilities
High demand, favorable cost	$22320 - 7440$	$26600 - 12600$	$0,50 \cdot 0,60$
	$= 14880$	$= 14000$	$= 0,30$
High demand, unfavorable cost	$22320 - 15500$	$26600 - 18200$	$0,50 \cdot 0,40$
	$= 6820$	$= 8400$	$= 0,20$
Low demand, favorable cost	$10800 - 3600$	$13680 - 6480$	$0,50 \cdot 0,60$
	$= 7200$	$= 7200$	$= 0,30$
Low demand, unfavorable cost	$10800 - 7500$	$13680 - 9360$	$0,50 \cdot 0,40$
	$= 3300$	$= 4320$	$= 0,20$

Combined probabilities are $0,50 \cdot 0,60 = 0,30$, $0,50 \cdot 0,40 = 0,20$, $0,50 \cdot 0,60 = 0,30$, $0,50 \cdot 0,40 = 0,20$.

State of nature	Probability	Model A	Model B
High demand, favorable cost	0.30	14880	14000
High demand, unfavorable cost	0.20	6820	8400
Low demand, favorable cost	0.30	7200	7200
Low demand, unfavorable cost	0.20	3300	4320

C4

$$\text{máx}(\text{Model A}) = \text{máx}\{14880, 6820, 7200, 3300\} = 14880,$$

$$\text{máx}(\text{Model B}) = \text{máx}\{14000, 8400, 7200, 4320\} = 14000.$$

$$\max\{\max(\text{Model A}), \max(\text{Model B})\} = \max\{14880, 14000\} = 14880.$$

The Maximax choice is Model A with 14880. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\min(\text{Model A}) = \min\{14880, 6820, 7200, 3300\} = 3300,$$

$$\min(\text{Model B}) = \min\{14000, 8400, 7200, 4320\} = 4320.$$

$$\max\{\min(\text{Model A}), \min(\text{Model B})\} = \max\{3300, 4320\} = 4320.$$

The Maximin choice is Model B with 4320 because it has the least severe worst-case loss. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined-state probabilities.

$$\begin{aligned} EV_A &= 0,30(14880) + 0,20(6820) + 0,30(7200) + 0,20(3300) \\ &= 4464 + 1364 + 2160 + 660 = 8648, \end{aligned}$$

$$\begin{aligned} EV_B &= 0,30(14000) + 0,20(8400) + 0,30(7200) + 0,20(4320) \\ &= 4200 + 1680 + 2160 + 864 = 8904. \end{aligned}$$

The expected value criterion chooses Model B because 8904 is the largest expected profit. This differs from the Maximax choice Model A and agrees with the Maximin choice Model B, so the expected value balances payoffs and probabilities.